

15. The analysis is similar to that given in the previous exercise, and the probability is 0.47.

16. In the usual notation, we have

$$\begin{aligned}\Pr(B_2) &= \Pr(A_1 \cap B_2) + \Pr(B_1 \cap B_2) = \Pr(A_1) \Pr(B_2 \mid A_1) + \Pr(B_1) \Pr(B_2 \mid B_1) \\ &= \frac{1}{4} \cdot \frac{2}{3} + \frac{3}{4} \cdot \frac{1}{3} = \frac{5}{12}.\end{aligned}$$

17. Clearly, we must assume that $\Pr(B_j \cap C) > 0$ for all j , otherwise (2.1.5) is undefined. By applying the definition of conditional probability to each term, the right side of (2.1.5) can be rewritten as

$$\sum_{i=1}^k \frac{\Pr(B_j \cap C)}{\Pr(C)} \frac{\Pr(A \cap B_j \cap C)}{\Pr(B_j \cap C)} = \frac{1}{\Pr(C)} \sum_{i=1}^k \Pr(A \cap B_j \cap C).$$

According to the law of total probability, the last sum above is $\Pr(A \cap C)$, hence the ratio is $\Pr(A|C)$.

2.2 Independent Events

Commentary

Near the end of this section, we introduce conditionally independent events. This is a prelude to conditionally independent and conditionally i.i.d. random variables that are introduced in Sec. 3.7. Conditional independence has become more popular in statistical modeling with the introduction of latent-variable models and expert systems. Although these models are not introduced in this text, students who will encounter them in the future would do well to study conditional independence early and often.

Conditional independence is also useful for illustrating how learning data can change the distribution of an unknown value. The first examples of this come in Sec. 2.3 after Bayes' theorem. The assumption that a sample of random variables is conditionally i.i.d. given an unknown parameter is the analog in Bayesian inference to the assumption that the random sample is i.i.d. marginally. Instructors who are not going to cover Bayesian topics might wish to bypass this material, even though it can also be useful in its own right. If you decide to not discuss conditional independence, then there is some material later in the book that you might wish to bypass as well:

- Exercise 23 in this section.
- The discussion of conditionally independent events on pages 81–84 in Sec. 2.3.
- Exercises 12, 14 and 15 in Sec. 2.3.
- The discussion of conditionally independent random variables that starts on page 163.
- Exercises 13 and 14 in Sec. 3.7.
- Virtually all of the Bayesian material.

This section ends with an extended example called “The Collector’s Problem”. This example combines methods from Chapters 1 and 2 to solve an easily stated but challenging problem.

Solutions to Exercises

1. If $\Pr(B) < 1$, then $\Pr(B^c) = 1 - \Pr(B) > 0$. We then compute

$$\begin{aligned}
 \Pr(A^c|B^c) &= \frac{\Pr(A^c \cap B^c)}{\Pr(B^c)} \\
 &= \frac{1 - \Pr(A \cup B)}{1 - \Pr(B)} \\
 &= \frac{1 - \Pr(A) - \Pr(B) + \Pr(A \cap B)}{1 - \Pr(B)} \\
 &= \frac{1 - \Pr(A) - \Pr(B) + \Pr(A) \Pr(B)}{1 - \Pr(B)} \\
 &= \frac{[1 - \Pr(A)][1 - \Pr(B)]}{1 - \Pr(B)} \\
 &= 1 - \Pr(A) = \Pr(A^c).
 \end{aligned}$$

2.

$$\begin{aligned}
 \Pr(A^c B^c) &= \Pr[(A \cup B)^c] = 1 - \Pr(A \cup B) \\
 &= 1 - [\Pr(A) + \Pr(B) - \Pr(A \cap B)] \\
 &= 1 - \Pr(A) - \Pr(B) + \Pr(A) \Pr(B) \\
 &= [1 - \Pr(A)][1 - \Pr(B)] \\
 &= \Pr(A^c) \Pr(B^c).
 \end{aligned}$$

3. Since the event $A \cap B$ is a subset of the event A , and $\Pr(A) = 0$, it follows that $\Pr(A \cap B) = 0$. Hence, $\Pr(A \cap B) = 0 = \Pr(A) \Pr(B)$.
4. The probability that the sum will be seven on any given roll of the dice is $1/6$. The probability that this event will occur on three successive rolls is therefore $(1/6)^3$.
5. The probability that both systems will malfunction is $(0.001)^2 = 10^{-6}$. The probability that at least one of the systems will function is therefore $1 - 10^{-6}$.
6. The probability that the man will win the first lottery is $100/10000 = 0.01$, and the probability that he will win the second lottery is $100/5000 = 0.02$. The probability that he will win at least one lottery is, therefore,

$$0.01 + 0.02 - (0.01)(0.02) = 0.0298.$$

7. Let E_1 be the event that A is in class, and let E_2 be the event that B is in class. Let C be the event that at least one of the students is in class. That is, $C = E_1 \cup E_2$.

- (a) We want $\Pr(C)$. We shall use Theorem 1.5.7 to compute the probability. Since E_1 and E_2 are independent, we have $\Pr(E_1 \cap E_2) = \Pr(E_1) \Pr(E_2)$. Hence

$$\Pr(C) = \Pr(E_1) + \Pr(E_2) - \Pr(E_1 \cap E_2) = 0.8 + 0.6 - 0.8 \times 0.6 = 0.92.$$

- (b) We want $\Pr(E_1|C)$. We computed $\Pr(C) = 0.92$ in part (a). Since $E_1 \subset C$, $\Pr(E_1 \cap C) = \Pr(E_1) = 0.8$. So, $\Pr(E_1|C) = 0.8/0.92 = 0.8696$.

8. The probability that all three numbers will be equal to a specified value is $1/6^3$. Therefore, the probability that all three numbers will be equal to any one of the six possible values is $6/6^3 = 1/36$.
9. The probability that exactly n tosses will be required on a given performance is $1/2^n$. Therefore, the probability that exactly n tosses will be required on all three performances is $(1/2^n)^3 = 1/8^n$. The probability that the same number of tosses will be required on all three performances is $\sum_{n=1}^{\infty} \frac{1}{8^n} = \frac{1}{7}$.
10. The probability p_j that exactly j children will have blue eyes is

$$p_j = \binom{5}{j} \left(\frac{1}{4}\right)^j \left(\frac{3}{4}\right)^{5-j} \quad \text{for } j = 0, 1, \dots, 5.$$

The desired probability is

$$\frac{p_3 + p_4 + p_5}{p_1 + p_2 + p_3 + p_4 + p_5}.$$

11. (a) We must determine the probability that at least two of the four oldest children will have blue eyes. The probability p_j that exactly j of these four children will have blue eyes is

$$p_j = \binom{4}{j} \left(\frac{1}{4}\right)^j \left(\frac{3}{4}\right)^{4-j}.$$

The desired probability is therefore $p_2 + p_3 + p_4$.

- (b) The two different types of information provided in Exercise 10 and part (a) are similar to the two different types of information provided in part (a) and part (b) of Exercise 9 of Sec. 2.1.

12. (a) $\Pr(A^c \cap B^c \cap C^c) = \Pr(A^c) \Pr(B^c) \Pr(C^c) = \frac{3}{4} \cdot \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{4}$.
- (b) The desired probability is

$$\begin{aligned} \Pr(A \cap B^c \cap C^c) + \Pr(A^c \cap B \cap C^c) + \Pr(A^c \cap B^c \cap C) &= \frac{1}{4} \cdot \frac{2}{3} \cdot \frac{1}{2} \\ &+ \frac{3}{4} \cdot \frac{1}{3} \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{2}{3} \cdot \frac{1}{2} = \frac{11}{24}. \end{aligned}$$

13. The probability of obtaining a particular sequence of ten particles in which one particle penetrates the shield and nine particles do not is $(0.01)(0.99)^9$. Since there are 10 such sequences in the sample space, the desired probability is $10(0.01)(0.99)^9$.
14. The probability that none of the ten particles will penetrate the shield is $(0.99)^{10}$. Therefore, the probability that at least one particle will penetrate the shield is $1 - (0.99)^{10}$.
15. If n particles are emitted, the probability that at least one particle will penetrate the shield is $1 - (0.99)^n$. In order for this value to be at least 0.8 we must have

$$\begin{aligned} 1 - (0.99)^n &\geq 0.8 \\ (0.99)^n &\leq 0.2 \\ n \log(0.99) &\leq \log(0.2). \end{aligned}$$

Since $\log(0.99)$ is negative, this final relation is equivalent to the relation

$$n \geq \frac{\log(0.2)}{\log(0.99)} \approx 160.1.$$

So 161 or more particles are needed.

16. To determine the probability that team A will win the World Series, we shall calculate the probabilities that A will win in exactly four, five, six, and seven games, and then sum these probabilities. The probability that A will win four straight game is $(1/3)^4$. The probability that A will win in five games is equal to the probability that the fourth victory of team A will occur in the fifth game. As explained in Example 2.2.8, this probability is $\binom{4}{3} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)$. Similarly, the probabilities that A will win in six games and in seven games are $\binom{5}{3} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^2$ and $\binom{6}{3} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^3$, respectively. By summing these probabilities, we obtain the result $\sum_{i=3}^6 \binom{i}{3} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^{i-3}$, which equals 379/2187.

A second way to solve this problem is to pretend that all seven games are going to be played, regardless of whether one team has won four games before the seventh game. From this point of view, of the seven games that are played, the team that wins the World Series might win four, five, six, or seven games. Therefore, the probability that team A will win the series can be determined by calculating the probabilities that team A will win exactly four, five, six, and seven games, and then summing these probabilities. In this way, we obtain the result

$$\sum_{i=4}^7 \binom{7}{i} \left(\frac{1}{3}\right)^i \left(\frac{2}{3}\right)^{7-i}.$$

It can be shown that this answer is equal to the answer that we obtained first.

17. In order for the target to be hit for the first time on the third throw of boy A , all five of the following independent events must occur: (1) A misses on his first throw, (2) B misses on his first throw, (3) A misses on his second throw, (4) B misses on his second throw, (5) A hits on his third throw. The probability of all five events occurring is $\frac{2}{3} \cdot \frac{3}{4} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{1}{3} = \frac{1}{12}$.
18. Let E denote the event that boy A hits the target before boy B . There are two methods of solving this problem. The first method is to note that the event E can occur in two different ways: (i) If A hits the target on the first throw. This event occurs with probability $\frac{1}{3}$. (ii) If both A and B miss the target on their first throws, and then subsequently A hits the target before B . The probability that A and B will both miss on their first throws is $\frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2}$. When they do miss, the conditions of the game become exactly the same as they were at the beginning of the game. In effect, it is as if the boys were starting a new game all over again, and so the probability that A will subsequently hit the target before B is again $\Pr(E)$. Therefore, by considering these two ways in which the event E can occur, we obtain the relation

$$\Pr(E) = \frac{1}{3} + \frac{1}{2} \Pr(E).$$

The solution is $\Pr(E) = \frac{2}{3}$.

The second method of solving the problem is to calculate the probabilities that the target will be hit for the first time on boy A 's first throw, on his second throw, on his third throw, etc., and then to sum these probabilities. For the target to be hit for the first time on his j th throw, both A and B must miss on each of their first $j - 1$ throws, and then A must hit on his next throw. The probability of this event is

$$\left(\frac{2}{3}\right)^{j-1} \left(\frac{3}{4}\right)^{j-1} \left(\frac{1}{3}\right) = \left(\frac{1}{2}\right)^{j-1} \left(\frac{1}{3}\right).$$

Hence,

$$\Pr(E) = \frac{1}{3} \sum_{j=1}^{\infty} \left(\frac{1}{2}\right)^{j-1} = \frac{1}{3}(2) = \frac{2}{3}.$$

19. Let A_1 denote the event that no red balls are selected, let A_2 denote the event that no white balls are selected, and let A_3 denote the event that no blue balls are selected. We must determine the value of $\Pr(A_1 \cup A_2 \cup A_3)$. We shall apply Theorem 1.10.1. The event A_1 will occur if and only if all ten selected balls are white or blue. Since there is probability 0.8 that any given selected ball will be white or blue, we have $\Pr(A_1) = (0.8)^{10}$. Similarly, $\Pr(A_2) = (0.7)^{10}$ and $\Pr(A_3) = (0.5)^{10}$. The event $A_1 \cap A_2$ will occur if and only if all ten selected balls are blue. Therefore $\Pr(A_1 \cap A_2) = (0.5)^{10}$. Similarly, $\Pr(A_2 \cap A_3) = (0.2)^{10}$ and $\Pr(A_1 \cap A_3) = (0.3)^{10}$. Finally, the event $A_1 \cap A_2 \cap A_3$ cannot possibly occur, so $\Pr(A_1 \cap A_2 \cap A_3) = 0$. So, the desired probability is

$$(0.8)^{10} + (0.7)^{10} + (0.5)^{10} - (0.5)^{10} - (0.2)^{10} - (0.3)^{10} \approx 0.1356.$$

20. To prove that $B_1, \dots, B_k$ are independent events, we must prove that for every subset of r of these events ($r = 1, \dots, k$), we have

$$\Pr(B_{i_1} \cap \dots \cap B_{i_r}) = \Pr(B_{i_1}) \cdots \Pr(B_{i_r}).$$

We shall simplify the notation by writing simply $B_1, \dots, B_r$ instead of $B_{i_1}, \dots, B_{i_r}$. Hence, we must show that

$$\Pr(B_1 \cap \dots \cap B_r) = \Pr(B_1) \cdots \Pr(B_r). \quad (\text{S.2.2})$$

Suppose that the relation (S.2.2) is satisfied whenever $B_j = A_j^c$ for m or fewer values of j and $B_j = A_j$ for the other $k - m$ or more values of j . We shall show that (S.2.2) is also satisfied whenever $B_j = A_j^c$ for $m + 1$ values of j . Without loss of generality, we shall assume that $j = r$ is one of these $m + 1$ values, so that $B_r = A_r^c$. It is always true that

$$\Pr(B_1 \cap \dots \cap B_r) = \Pr(B_1 \cap \dots \cap B_{r-1}) - \Pr(B_1 \cap \dots \cap B_{r-1} \cap B_r^c).$$

Since among the events $B_1, \dots, B_{r-1}$ there are m or fewer values of j such that $B_j = A_j^c$, it follows from the induction hypothesis that

$$\Pr(B_1 \cap \dots \cap B_{r-1}) = \Pr(B_1) \cdots \Pr(B_{r-1}).$$

Furthermore, since $B_r^c = A_r$, the same induction hypothesis implies that

$$\Pr(B_1 \cap \dots \cap B_{r-1} B_r^c) = \Pr(B_1) \cdots \Pr(B_{r-1}) \Pr(B_r^c).$$

It now follows that

$$\Pr(B_1 \cap \dots \cap B_r) = \Pr(B_1) \cdots \Pr(B_{r-1})[1 - \Pr(B_r^c)] = \Pr(B_1) \cdots \Pr(B_r).$$

Thus, we have shown that if the events $B_1, \dots, B_k$ are independent whenever there are m or fewer values of j such that $B_j = A_j^c$, then the events $B_1, \dots, B_k$ are also independent whenever there are $m + 1$ values of j such that $B_j = A_j^c$. Since $B_1, \dots, B_k$ are obviously independent whenever there are zero values of j such that $B_j = A_j^c$ (i.e., whenever $B_j = A_j$ for $j = 1, \dots, k$), the induction argument is complete. Therefore, the events $B_1, \dots, B_k$ are independent regardless of whether $B_j = A_j$ or $B_j = A_j^c$ for each value of j .

21. For the “only if” direction, we need to prove that if $A_1, \dots, A_k$ are independent then

$$\Pr(A_{i_1} \cap \dots \cap A_{i_m} | A_{j_1} \cap \dots \cap A_{j_\ell}) = \Pr(A_{i_1} \cap \dots \cap A_{i_m}),$$

for all disjoint subsets $\{i_1, \dots, i_m\}$ and $\{j_1, \dots, j_\ell\}$ of $\{1, \dots, k\}$. If $A_1, \dots, A_k$ are independent, then

$$\Pr(A_{i_1} \cap \dots \cap A_{i_m} \cap A_{j_1} \cap \dots \cap A_{j_\ell}) = \Pr(A_{i_1} \cap \dots \cap A_{i_m}) \Pr(A_{j_1} \cap \dots \cap A_{j_\ell}),$$

hence it follows that

$$\Pr(A_{i_1} \cap \dots \cap A_{i_m} | A_{j_1} \cap \dots \cap A_{j_\ell}) = \frac{\Pr(A_{i_1} \cap \dots \cap A_{i_m} \cap A_{j_1} \cap \dots \cap A_{j_\ell})}{\Pr(A_{j_1} \cap \dots \cap A_{j_\ell})} = \Pr(A_{i_1} \cap \dots \cap A_{i_m}).$$

For the “if” direction, assume that $\Pr(A_{i_1} \cap \dots \cap A_{i_m} | A_{j_1} \cap \dots \cap A_{j_\ell}) = \Pr(A_{i_1} \cap \dots \cap A_{i_m})$ for all disjoint subsets $\{i_1, \dots, i_m\}$ and $\{j_1, \dots, j_\ell\}$ of $\{1, \dots, k\}$. We must prove that $A_1, \dots, A_k$ are independent. That is, we must prove that for every subset $\{s_1, \dots, s_n\}$ of $\{1, \dots, k\}$, $\Pr(A_{s_1} \cap \dots \cap A_{s_n}) = \Pr(A_{s_1}) \cdots \Pr(A_{s_n})$. We shall do this by induction on n . For $n = 1$, we have that $\Pr(A_{s_1}) = \Pr(A_{s_1})$ for each subset $\{s_1\}$ of $\{1, \dots, k\}$. Now, assume that for all $n \leq n_0$ and for all subsets $\{s_1, \dots, s_n\}$ of $\{1, \dots, k\}$ it is true that $\Pr(A_{s_1} \cap \dots \cap A_{s_n}) = \Pr(A_{s_1}) \cdots \Pr(A_{s_n})$. We need to prove that for every subset $\{t_1, \dots, t_{n_0+1}\}$ of $\{1, \dots, k\}$

$$\Pr(A_{t_1} \cap \dots \cap A_{t_{n_0+1}}) = \Pr(A_{t_1}) \cdots \Pr(A_{t_{n_0+1}}). \quad (\text{S.2.3})$$

It is clear that

$$\Pr(A_{t_1} \cap \dots \cap A_{t_{n_0+1}}) = \Pr(A_{t_1} \cap \dots \cap A_{t_{n_0}} | A_{t_{n_0+1}}) \Pr(A_{t_{n_0+1}}). \quad (\text{S.2.4})$$

We have assumed that $\Pr(A_{t_1} \cap \dots \cap A_{t_{n_0}} | A_{t_{n_0+1}}) = \Pr(A_{t_1} \cap \dots \cap A_{t_{n_0}})$ for all disjoint subsets $\{t_1, \dots, t_{n_0}\}$ and $\{t_{n_0+1}\}$ of $\{1, \dots, k\}$. Since the right side of this last equation is the probability of the intersection of only n_0 events, then we know that

$$\Pr(A_{t_1} \cap \dots \cap A_{t_{n_0}}) = \Pr(A_{t_1}) \cdots \Pr(A_{t_{n_0}}).$$

Combining this with Eq. (S.2.4) implies that (S.2.3) holds.

22. For the “only if” direction, we assume that A_1 and A_2 are conditionally independent given B and we must prove that $\Pr(A_2 | A_1 \cap B) = \Pr(A_2 | B)$. Since A_1 and A_2 are conditionally independent given B , $\Pr(A_1 \cap A_2 | B) = \Pr(A_1 | B) \Pr(A_2 | B)$. This implies that

$$\Pr(A_2 | B) = \frac{\Pr(A_1 \cap A_2 | B)}{\Pr(A_1 | B)}.$$

Also,

$$\Pr(A_2|A_1 \cap B) = \frac{\Pr(A_1 \cap A_2 \cap B)}{\Pr(A_1 \cap B)} = \frac{\Pr(A_1 \cap A_2 \cap B)/\Pr(B)}{\Pr(A_1 \cap B)/\Pr(B)} = \frac{\Pr(A_1 \cap A_2|B)}{\Pr(A_1|B)}.$$

Hence, $\Pr(A_2|A_1 \cap B) = \Pr(A_2|B)$.

For the “if” direction, we assume that $\Pr(A_2|A_1 \cap B) = \Pr(A_2|B)$, and we must prove that A_1 and A_2 are conditionally independent given B . That is, we must prove that $\Pr(A_1 \cap A_2|B) = \Pr(A_1|B) \Pr(A_2|B)$. We know that

$$\Pr(A_1 \cap A_2|B) \Pr(B) = \Pr(A_2|A_1 \cap B) \Pr(A_1 \cap B),$$

since both sides are equal to $\Pr(A_1 \cap A_2 \cap B)$. Divide both sides of this equation by $\Pr(B)$ and use the assumption $\Pr(A_2|A_1 \cap B) = \Pr(A_2|B)$ together with $\Pr(A_1 \cap B)/\Pr(B) = \Pr(A_1|B)$ to obtain

$$\Pr(A_1 \cap A_2|B) = \Pr(A_2|B) \Pr(A_1|B).$$

23. (a) Conditional on B the events $A_1, \dots, A_{11}$ are independent with probability 0.8 each. The conditional probability that a particular collection of eight programs out of the 11 will compile is $0.8^8 0.2^3 = 0.001342$. There are $\binom{11}{8} = 165$ different such collections of eight programs out of the 11, so the probability of exactly eight programs will compile is $165 \times 0.001342 = 0.2215$.
- (b) Conditional on B^c the events $A_1, \dots, A_{11}$ are independent with probability 0.4 each. The conditional probability that a particular collection of eight programs out of the 11 will compile is $0.4^8 0.6^3 = 0.0001416$. There are $\binom{11}{8} = 165$ different such collections of eight programs out of the 11, so the probability of exactly eight programs will compile is $165 \times 0.0001416 = 0.02335$.
24. Let $n > 1$, and assume that $A_1, \dots, A_n$ are mutually exclusive. For the “if” direction, assume that at most one of the events has strictly positive probability. Then, the intersection of every collection of size 2 or more has probability 0. Also, the product of every collection of 2 or more probabilities is 0, so the events satisfy Definition 2.2.2 and are mutually independent. For the “only if” direction, assume that the events are mutually independent. The intersection of every collection of size 2 or more is empty and must have probability 0. Hence the product of the probabilities of every collection of size 2 or more must be 0. This means that at least one factor from every product of at least 2 probabilities must itself be 0. Hence there can be no more than one of the probabilities greater than 0, otherwise the product of the two nonzero probabilities would be nonzero.

2.3 Bayes’ Theorem

Commentary

This section ends with two extended discussions on how Bayes’ theorem is applied. The first involves a sequence of simple updates to the probability of a specific event. It illustrates how conditional independence allows one to use posterior probabilities after observing some events as prior probabilities before observing later events. This idea is subtle, but very useful in Bayesian inference. The second discussion builds upon this idea and illustrates the type of reasoning that can be used in real inference problems. Examples 2.3.7 and 2.3.8 are particularly useful in this regard. They show how data can bring very different prior probabilities into